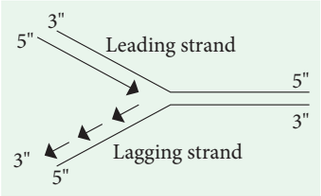
synthesized in short pieces called **Okazaki fragments**. In discontinuous strand where the Okazaki fragments are united by ligase is called Lagging strand where the replication direction is 5' → 3' which is opposite to the direction of fork movement. . The continuous strand is called Leading strand where the replication direction is 5' → 3' which is same to the direction to that of the replication fork movement. DNA ligase joins any nicks in the DNA by forming a phosphodiester bond between 3' hydroxyl and 5' phosphate group.



Arabidopsis telomere sequence - TTTAGGG

Plants Lacks Telomere Clock

Plant meristematic cells produces telomerase. The meristematic cells has an unlimited ability to divide. You have already studied about the telomeres in Chapter 6 and 8 of Class XI. In plants telomeres do not shrink as in somatic cells of vertebrates. Telomerase levels are higher in root tips and seedlings (renewable tissue) which has a higher amount of meristematic cells than proliferative structures like leaves.

What is the special mechanism which replicates chromosomal ends?

After the replication of the chromosomes, the enzyme **telomerase** adds several more repeats of DNA sequences to the telomeres. Telomerase use short RNA molecules as a template and add repeat sequences on to telomeres (DNA nucleotide polymerisation).

The Energetics of DNA Replication - Deoxyribonucleotides such as deoxyadenosine triphosphate dATP, dGTP, dCTP and dTTP provide energy for the synthesis of DNA. Purpose of Deoxyribonucelotides (1) acts as a substrate (2) provide energy for polymerisation.

3.6.2 Experimental evidence of DNA replication: Taylors Experiment

J. Herbert Taylor, Philip Woods and Walter Hughes demonstrated the semiconservative replication of DNA in the root cells of *Vicia faba*. They labelled DNA with ^{3}H Thymidine, a radioactive precursor of DNA and performed autoradiography. They grew root tips in a medium in the presence of radioactive labelled thymidine, so that the radioactivity was incorporated into the DNA of these cells. The outline of this labelled chromosomes appears in the form of scattered black dots of silver grains on a photographic film.

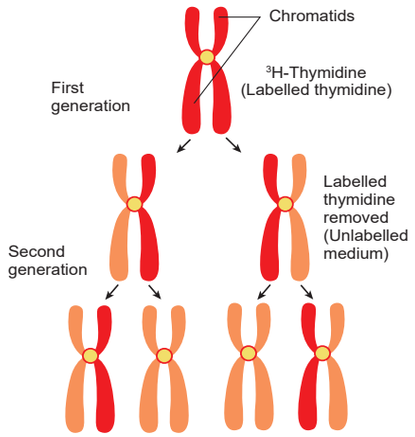


Figure: 3.19 Taylors Experiment on *Vicia faba*

The root tips with labelled chromosomes were placed in an unlabelled medium containing colchicine to arrest the culture at the metaphase and examine the chromosome by autoradiography. The observations were,

1. In the chromosome of **first generation the radioactivity** was found to be distributed to **both the chromatids** because in the original strand of DNA double helix was labelled with radioactivitiy and the new strand was unlabelled.
2. In the chromosome of the **second generation only one of the two chromatids** in each chromosome was radioactive (labelled).

The results proved the semiconservative method of DNA replication.

3.7 Protein synthesis in plants

The process of protein synthesis consists of two major steps, they are Transcription and Translation.

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